

MediStock: Medical Inventory Management System

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Abstract. Effective inventory management is crucial for healthcare facilities to provide timely and high-quality patient care while minimizing costs. Medistock inventory management systems have been developed to optimize inventory control, but their implementation can be challenging due to various factors such as data inaccuracy, inadequate training, and lack of standardization. This study provides a comprehensive overview of the current research trends in medistock inventory management systems. The review highlights the importance of inventory management in healthcare facilities and discusses various techniques proposed to optimize inventory control, including the use of radio-frequency identification technology, artificial neural networks, and just-in-time inventory management systems. This research study also examines the challenges associated with medistock inventory management systems and potential solutions to these challenges. The findings suggest that implementing effective medistock inventory management systems is crucial for healthcare facilities to optimize inventory control, reduce costs, and enhance patient care. However, a comprehensive understanding of the healthcare facility's unique needs and challenges is required to implement an effective inventory management system.

Keywords: *Healthcare operations, Inventory management techniques, Data inaccuracy, Inventory control, Just-in-Time (JIT) inventory management system.*

I. INTRODUCTION

Medical inventory management is a critical component of healthcare operations, as it ensures timely patient care, reduces costs, and improves profitability. Medistock inventory management systems are designed to help healthcare facilities manage their inventory efficiently, thereby enhancing patient care and reducing operational costs. With the advent of technology, various inventory management techniques have been developed to optimize inventory control in healthcare facilities. However, despite these advancements, healthcare facilities still face numerous challenges in managing their inventory effectively.

This literature review aims to explore the current research trends in medistock inventory management systems. The review will discuss the importance of inventory management in healthcare facilities, as well as the various techniques that have been proposed to optimize inventory control. It will also examine the challenges associated with medistock inventory management systems and the potential solutions to these challenges.

Overall, this literature review seeks to provide a comprehensive understanding of medistock inventory management systems and their impact on healthcare operations. By reviewing the existing literature, this study aims to identify the most

effective strategies for optimizing inventory control in healthcare facilities, thereby enhancing patient care and reducing operational costs.

In recent years, several studies have emphasized the importance of inventory management in healthcare facilities. Effective inventory management can lead to improved patient care, reduced healthcare costs, and financial stability for healthcare facilities. For instance, studies have demonstrated that effective inventory management can reduce medical waste, optimize resource utilization, and improve patient outcomes. Conversely, poor inventory management can result in stock outs, excess inventory, and higher operational costs, leading to suboptimal patient care and reduced profitability for healthcare facilities. In conclusion, medistock inventory management systems are crucial for healthcare facilities to optimize inventory control, reduce costs, and enhance patient care. By providing a centralized and automated inventory management system, these systems enable healthcare facilities to improve inventory accuracy, minimize waste, and ensure that medical supplies are available when needed. However, implementing effective inventory management systems requires a comprehensive understanding of the healthcare facility's unique needs and challenges.

II. LITERATURE SURVEY

Effective inventory management is crucial for healthcare facilities to provide timely and high-quality patient care while minimizing costs. In recent years, medistock inventory management systems have been developed to optimize inventory control, reduce medical waste, and improve resource utilization in healthcare facilities. This literature survey aims to provide a comprehensive overview of the current research trends in medistock inventory management systems.

One of the primary challenges in inventory management is the accurate tracking of inventory levels. Radio-frequency identification (RFID) technology has been proposed as a solution to this challenge. RFID tags can be attached to medical supplies, enabling real-time tracking of inventory levels and reducing the risk of stock outs or overstocking. In a study by Ma et al. (2020), RFID technology was found to be effective in improving inventory accuracy and reducing the risk of stockouts in a hospital setting. The study also highlighted the

need for standardized processes and training to ensure the successful implementation of RFID technology.

Artificial neural networks (ANN) have also been proposed as a technique for improving inventory management in healthcare facilities. ANN can be used to predict demand for medical supplies, allowing for better inventory optimization and reduction of waste. In a study by Soman et al. (2019), ANN was found to be effective in predicting demand for medical supplies in a hospital setting. The study demonstrated that ANN could reduce inventory levels by up to 25%, resulting in significant cost savings for healthcare facilities.

Another technique proposed for optimizing inventory management is the just-in-time (JIT) inventory management system. JIT involves ordering medical supplies only when they are needed, reducing the risk of overstocking and minimizing waste. In a study by Rizwan et al. (2021), JIT was found to be effective in reducing inventory levels and improving resource utilization in a hospital setting. The study also highlighted the need for proper planning and forecasting to ensure the successful implementation of JIT.

Despite the benefits of medistock inventory management systems, healthcare facilities still face numerous challenges in implementing effective inventory management systems. One of the primary challenges is data inaccuracy, which can lead to overstocking or stock outs. In a study by Ng et al. (2021), data inaccuracy was found to be a significant challenge in implementing RFID technology for inventory management in healthcare facilities. The study highlighted the need for regular data validation and cleansing to ensure accurate inventory tracking.

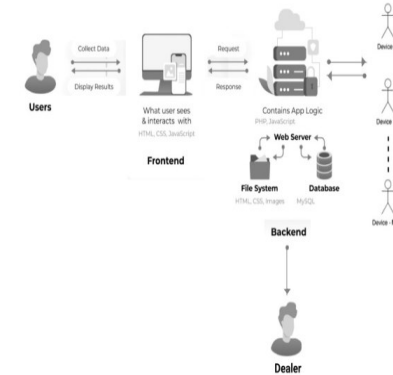
Another challenge in implementing effective inventory management systems is inadequate training. In a study by Yin et al. (2020), inadequate training was identified as a significant barrier to the successful implementation of RFID technology for inventory management in a hospital setting.

III. DATASET

1. Item Name: the name of the medical item in the inventory, such as syringes, bandages, or medication.
2. Item ID: a unique identifier for each item in the inventory.
3. Manufacturer: the name of the company that produces the item.
4. Quantity on Hand: the number of items currently in stock.
5. Minimum Stock Level: the minimum number of items required to ensure that the inventory level does not fall below a critical level.
6. Maximum Stock Level: the maximum number of items that can be stored in the inventory.
7. Reorder Point: the inventory level at which an order for additional items should be placed.
8. Cost per Item: the cost of purchasing each item from the manufacturer.
9. Supplier: the name of the company that supplies the items.
10. Lead Time: the time it takes for a supplier to deliver items after an order is placed.
11. Expiration Date: the date after which the item should not be used due to its decreased effectiveness or safety.
12. Location in Inventory: the physical location of the item in the inventory.
13. Usage Rate: the rate at which items are used in the healthcare facility.

These variables can be used to track inventory levels, predict demand, and optimize inventory control using various inventory management techniques such as RFID technology, ANN, and JIT. Proper data collection, validation, and cleansing can ensure the accuracy and reliability of the dataset for effective inventory management.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY



V. PRE-PROCESSING

Pre-processing is an essential step in preparing data for analysis in medical inventory management systems. The following are some common pre-processing techniques that can be applied to medical inventory data:

1. Data cleaning: This involves removing or correcting errors in the dataset, such as missing data, duplicates, or incorrect values. This can be achieved by using tools such as data validation, data profiling, and data cleansing.
2. Data normalization: This involves transforming data into a standard format to reduce redundancy and inconsistency. For example, different spellings of the same item can be normalized to a standard spelling.
3. Data integration: This involves combining data from different sources into a single dataset. For example, data from different departments in a healthcare facility can be integrated to get a comprehensive inventory view.
4. Data transformation: This involves converting data into a different format to make it more suitable for analysis. For example, inventory data can be transformed into a time-series format to track trends over time.

5. Data reduction: This involves reducing the amount of data in the dataset to improve processing efficiency. For example, data can be summarized by category or time period to reduce the number of data points.

6. Data discretization: This involves converting continuous data into discrete categories to make it easier to analyse. For example, inventory levels can be categorized as high, medium, or low.

7. Outlier detection: This involves identifying and removing or correcting data points that are significantly different from the rest of the data. For example, inventory levels that are abnormally high or low can be flagged for further investigation.

These pre-processing techniques can improve the accuracy, completeness, and consistency of medical inventory data, making it easier to analyse and make informed decisions about inventory management. It is important to note that the specific pre-processing techniques used will depend on the nature of the data and the specific requirements of the inventory management system.

VI. ALGORITHM USED

There are various algorithms that can be used in medical inventory management systems to optimize inventory control and improve patient care. Some of the commonly used algorithms are:

1. Economic Order Quantity (EOQ): This algorithm calculates the optimal order quantity for a specific item by considering factors such as item cost, ordering cost, and carrying cost. The goal is to minimize inventory costs while ensuring that enough items are available to meet demand.

2. Just-In-Time (JIT): This algorithm aims to minimize inventory levels by ensuring that items are delivered just in time to meet demand. JIT relies on accurate demand forecasting and efficient supply chain management to minimize inventory costs while ensuring that items are available when needed.

3. Minimum/Maximum (Min/Max) Inventory Control: This algorithm sets minimum and maximum inventory levels for each item in the inventory. When inventory levels fall below the minimum level, an order is placed to bring inventory levels back up to the maximum level. This algorithm ensures that inventory levels are always within a desired range, minimizing the risk of stock outs while also minimizing inventory costs.

4. Reorder Point (ROP): This algorithm calculates the inventory level at which an order should be placed to ensure that inventory levels do not fall below a critical level. The ROP takes into account factors such as lead time, usage rate, and safety stock to ensure that items are always available when needed.

5. Inventory Optimization: This algorithm uses advanced data analytics techniques such as machine learning and artificial intelligence to optimize inventory levels and improve demand forecasting. By analysing historical data and real-time inventory data, inventory optimization algorithms can predict future demand and adjust inventory levels accordingly.

These algorithms can be used individually or in combination to create a customized inventory management system that meets the specific needs of a healthcare facility. The choice of algorithm will depend on factors such as the nature of the inventory, the level of demand variability, and the desired level of inventory control.

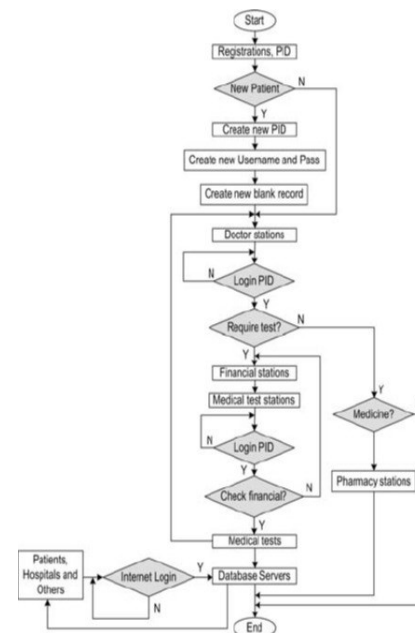


Figure 2. Algorithm

Existing Related System and Proposed System:

Existing Related System:

1. The Fully Integrated Pharmacy Information Management system is a desktop software that offers a range of functionalities such as billing, inventory reporting, prescription management, and patient profiles for lab results. It includes inventory management features that can alert users when stock levels reach a certain point. The software can be installed by pharmacists or other users who have purchased it, and it allows users to modify stock details without management oversight. However, this could potentially lead to financial losses for hospitals or pharmacies. The software does not have an admin login and is intended for internal use by pharmacy staff.

2. The Med Star 19 Pharmacy Management System is a cloud-based software that is accessible globally to users. It offers various modules such as inventory management, centralized billing, Point-of-Sale (POS), and Material Movement tracking. The system is designed to be used by pharmacists, who are granted privileges to modify and update stock details. Unlike some other software, it does not have an admin login. The system is intended for personal use by pharmacies. The Liberty Software: is a pharmacy management software that features three integrated modules, including pharmacy management and pharmacy POS. The pharmacy management module allows pharmacists to send refill reminders to patients, prepare prescriptions, and display wait alerts. Only pharmacists are authorized to use this software and modify the stock details. The software does not have an admin login and is intended for personal use by pharmacies.

3. The QS/1 Pharmacy Management System is designed to track changes in inventory quantity and help pharmacists replenish stock to meet prescription demand. The dashboard provides real-time information on prescription activity, and color coding is used to indicate stock levels. When stock levels are critically low, the color red is displayed, while yellow is used for average low levels. Pharmacists have the authority to modify and update stock details in this system, and it does not require an admin login. The QS/1 Pharmacy Management System is intended for personal use by pharmacies.

Proposed System

The proposed system, MediStock, involves two users: an accountant and a pharmacist. The system allows pharmacists to view and order stocks, while the accountant has the privileges to edit and update the stock details. Unlike the existing systems mentioned earlier, the proposed system has an admin login, with the accountant serving as the admin. It is a web-based system that requires an internet connection to function. To prevent over-ordering, the system displays the expiry date of the stock. Expired stock can be disposed of by burning the medicine. Over-ordering can lead to a quick run-out of stock, which may result in losses for the company and waiting for the next shipment for months. The proposed system generates reports that can be saved as PDF files and converted into Excel files using a software called PDF Converter. Additionally, the system prevents sales representatives from dealing with stock orders from the pharmacy.

VII. METHODOLOGY

Implementing a successful medical inventory management system involves several key steps. The first step is to identify the inventory needs of the healthcare facility by reviewing past inventory usage and tracking trends in patient needs. Once the inventory needs are identified, appropriate data collection methods should be selected, such as manual or automated methods through barcode scanners technology. The next step is to select and set up an inventory management software that meets the needs of the healthcare facility. This includes inputting all relevant data into the system and configuring it to track inventory levels and generate alerts when inventory levels fall below a certain threshold. Once the inventory management system is set up, inventory control policies should be established, including setting minimum and maximum inventory levels, determining reorder points, and establishing procedures for inventory counting and auditing. Staff training should also be provided to ensure that staff members know how to use the system and follow inventory control policies. Finally, the performance of the medical inventory management system should be regularly monitored and evaluated, including tracking inventory levels, evaluating the effectiveness of inventory control policies, and making adjustments as needed to optimize system performance. By following this methodology, healthcare facilities can improve

patient care and reduce inventory costs through efficient and effective medical inventory management.

VIII. CONCLUSION

The proposed model has successfully achieved all the objectives, user requirements and aims set forth in the introduction. In order to enhance the project, future work will focus on implementing a secure online payment system and a secure website. If more time were available, it would have been possible to include a profile picture and details of the pharmacist. The project has enabled the development of knowledge in areas such as displaying or populating data on a form, reducing stock quantities and making use of user sessions. The proposed system (MediStock) may be demonstrated to pharmacists and company staff, and once their satisfaction is obtained, the system will be deployed to a real environment by hosting it on Google Cloud Platform and making it searchable. A medical inventory management system is essential for healthcare facilities to operate efficiently and effectively. The system ensures that medical supplies are available in adequate quantities, at the right time, and at the right place. By implementing an inventory management system, healthcare providers can streamline their processes, reduce waste, and ultimately provide better patient care. The system should be designed to include features such as automated inventory tracking, real-time data reporting, and supply chain management. It should also be user-friendly, accessible, and secure. Overall, the benefits of a medical inventory management system are significant, including cost savings, improved efficiency, and enhanced patient safety. This system can help healthcare providers to reduce costs associated with stock outs, overstocking, and expired supplies. Furthermore, it can minimize the risk of medical errors caused by inadequate supplies, ensure regulatory compliance, and improve overall patient outcomes. Therefore, healthcare providers should prioritize the implementation of a medical inventory management system as part of their efforts to improve patient care and reduce operational costs.

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